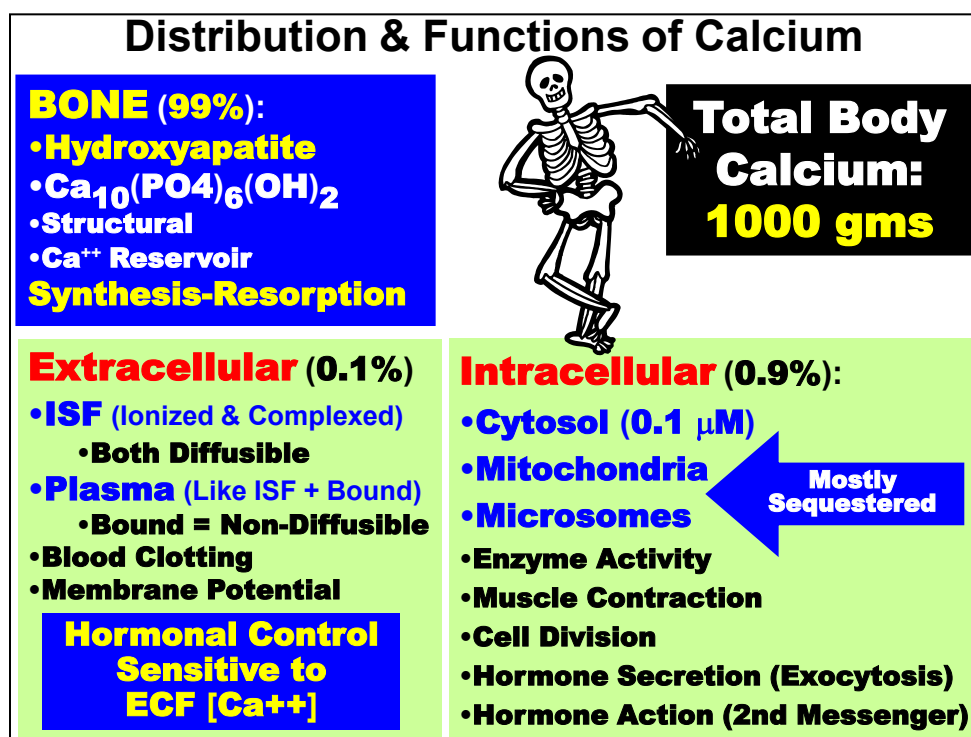


Narrated PowerPoint Presentations

- 1. Overview & Principles of Endocrine System**
- 2. Hypothalamic Neuroendocrine Hormones**
- 3. Anterior Pituitary Hormones**
- 4. Posterior Pituitary Hormones**
- 5. Thyroid Hormones Basic Properties**
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- 11. Adrenal Medulla Hormones**



The adult human body possess ≈ 1000 gms of **Calcium**, which is mostly distributed in **Bone**. However, smaller but functionally important **Extracellular & Intracellular** pools also exist.

Bone Calcium (top left text box).

About **99%** of total body calcium is contained in bone, primarily in the form of **Hydroxyapatite** crystals ($\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$).

Bone primarily serves to maintain **Skeletal Structure**, but also functions as a large **Ca^{++} Reservoir** that is stored (**Bone Synthesis**) or mobilized (**Bone Resorption**) in response to deviations in extracellular & intracellular **Ca^{++}** levels. Thus, bone plays an important role in calcium homeostasis.

Extracellular Fluid Calcium (bottom left text box).

About **0.1%** of total calcium exists extracellularly in **Interstitial Fluids (ISF) & Plasma**. Extracellular fluid calcium is vital for **Blood Clotting** and maintenance of the normal electrochemical **Membrane Potential**.

Interstitial Fluid Calcium ($1.5 \times 10^{-3} \text{ M}$) exists mainly as **Ionized Ca^{++}** although some is **Complexed** by anions (e.g. citrate, lactate, phosphate). Both forms of **ISF** calcium freely Diffuse through capillary membranes and equilibrate with **Ca^{++}** pools in **Plasma** as well as a soluble calcium pool in bone.

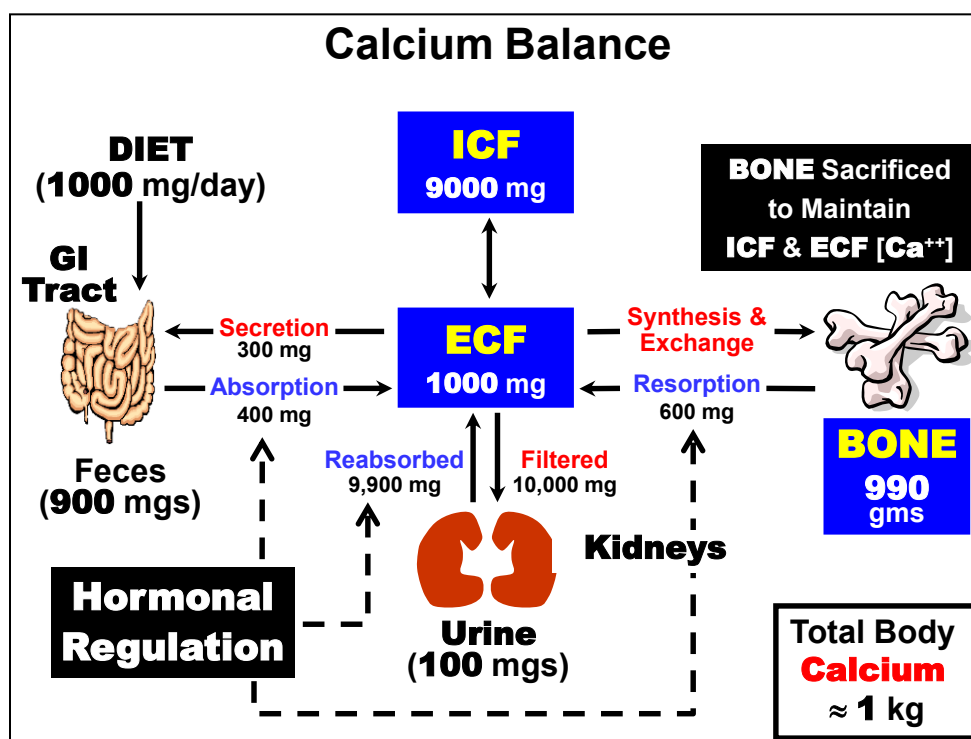
Plasma Calcium ($2.5 \times 10^{-3} \text{ M}$) also exists as **Ionized & Complexed Ca^{++}** . However, plasma calcium is also significantly **Bound** to plasma proteins (albumins & globulins), which cannot diffuse across capillary membranes. This **Non-Diffusible** calcium accounts for the higher **$[\text{Ca}^{++}]$** in **plasma** relative to **ISF**.

Intracellular Calcium (bottom right text box). Intracellular calcium is involved in the regulation of many important cellular processes, some of which are listed.

Approximately **0.9%** of total body calcium is contained intracellularly in soft tissue cells. However, intracellular calcium is **Mostly Sequestered** within **Mitochondria & Microsomes**.

Only a small concentration of **Ca^{++}** is actually maintained in the **Cytoplasm**, which is equilibrated with **ECF** ionized **Ca^{++}** pools. An intracellular **Bound** calcium fraction also exists.

NOTE: An important concept related to calcium regulation is that **Hormonal Control** mechanisms are primarily sensitive to deviations in **Extracellular $[\text{Ca}^{++}]$** (primarily **Plasma $[\text{Ca}^{++}]$**). Fluctuations in intracellular & bone calcium pools are regulated indirectly through their equilibration with the regulated **ECF Ca^{++}** pool.



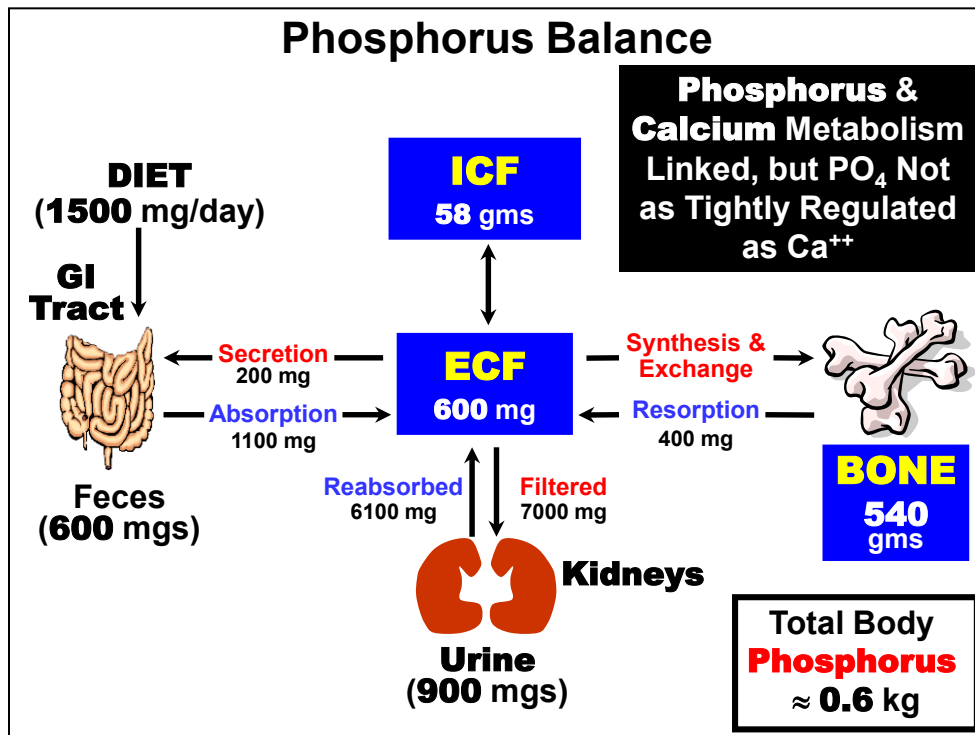
Maintenance of body Ca^{++} pools in the **ICF** (top center), **ECF** (center), & **Bone** (right) compartments occurs in coordination with the absorptive & secretory actions of the **GI Tract** (left) and the filtration, reabsorptive & excretory functions of the **Kidneys** (bottom).

The average daily U.S. intake of calcium in the diet (top left) is ≈ 1000 mg/day (8-oz of milk = 290 mg calcium). Calcium absorbed from the GI Tract exchanges with and replenishes the various body pools (Bone, ICF, ECF). Excesses are ultimately lost in the Urine & Feces. Normal adults maintain a net Calcium Balance.

That is, daily intake from the diet (1000 mg) equals daily loss in urine (100 mgs) & feces (900 mgs) such that steady-state levels within compartments are maintained.

Calcium equilibrium is maintained through **Hormonal Regulation of Intestinal Absorption, Kidney Reabsorption, & Bone Resorption & Synthesis** (dashed arrows).

NOTE: An important principle related to calcium homeostasis is that **Bone Calcium Will Always Be Sacrificed to Maintain ICF & ECF [Ca⁺⁺]**. This property of calcium homeostasis is a major reason for conditions such as Osteoporosis and other bone weakening disorders related to a hypocalcemic state.



Phosphorus is required for various and important physiological functions. In addition to its structural role, phosphorus is also a component of Nucleic Acids and high-energy metabolites such as ATP. Phosphorus is indispensable for signal transduction (cAMP & IP3) and is the terminal covalent source for various regulatory Kinases.

Calcium & Phosphorus metabolism and regulation are closely linked.

Normal adults possess a total of ≈ 600 grams of phosphorus. About 85-90% (540 gms) is found complexed with calcium in hydroxyapatite bone crystals. In addition, a lower amount (58 gms) exists in intracellular fluid or is incorporated cellularly into tissue cell membranes & cell components (combined = ICF Ca^{++}). The remainder (600 mg) is found extracellularly (ECF) either as free ion, in complex with cations (e.g. calcium), or bound to plasma proteins.

As summarized above, maintenance of daily phosphorus balance is also similar to that of calcium. Phosphorus intake is ≈ 1500 mg/day. Balance is maintained through regulated actions of the GI Tract & Kidneys.

An important difference is that although hormonal regulation occurs, **Phosphate is Not as Tightly Controlled Hormonally compared to calcium**. Consequently, phosphorus level varies more under influences such as diet, age, & gender.

Hormonal Control of Ca^{++} & PO_3^- Homeostasis

PARATHYROID HORMONE:

- **Increases** Plasma $[\text{Ca}^{++}]$
- **Decreases** Plasma $[\text{PO}_3^-]$

1,25 (OH) $_2$ -CHOLECALCIFEROL:

**VITAMIN D
(CALCITRIOL)**

- **Increases** Plasma $[\text{Ca}^{++}]$
- **Increases** Plasma $[\text{PO}_3^-]$

CALCITONIN:

- **Decreases** Plasma $[\text{Ca}^{++}]$ & $[\text{PO}_3^-]$

Three hormones are primarily involved in the regulation of calcium & phosphorus balance:

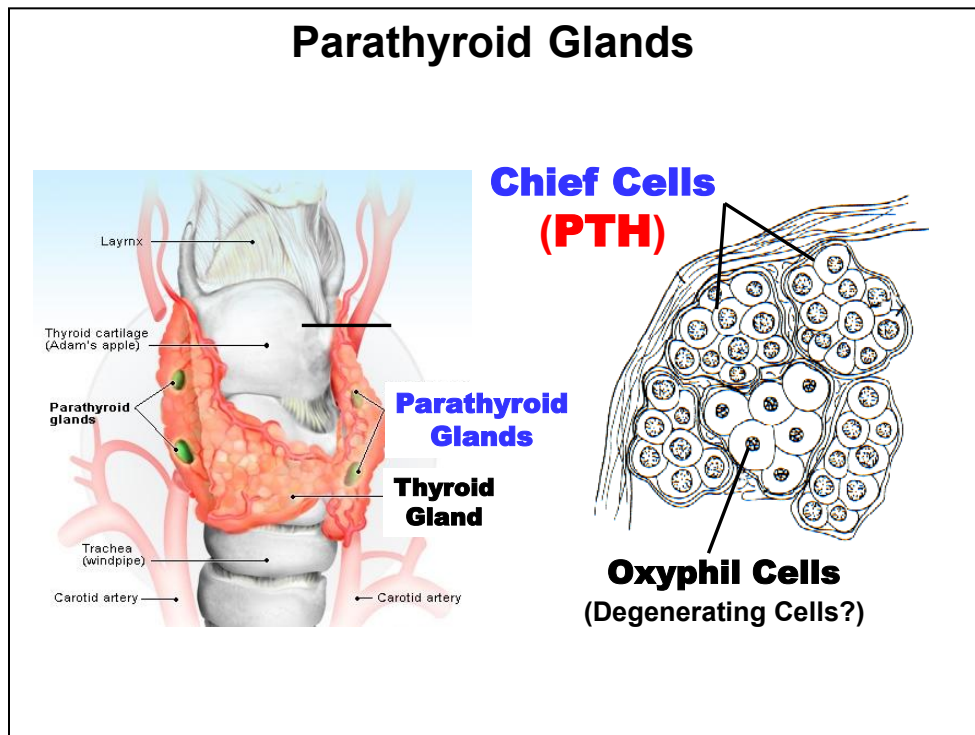
Parathyroid hormone (PTH) is a **Protein Hormone** secreted by the **Parathyroid Glands**. PTH acts directly on Bone and the Renal Tubules and indirectly on the GI Tract to Increase plasma $[\text{Ca}^{++}]$ and to Decrease plasma $[\text{PO}_3^-]$.

1,25(OH) $_2$ -Cholecalciferol (Vitamin D; Calcitriol) is a Steroid-like (Sterol) Hormone that acts mainly on the GI Tract, and less potently on Bone and the Renal Tubules to Increase plasma $[\text{Ca}^{++}]$ and $[\text{PO}_3^-]$.

Calcitonin is a **Protein Hormone** secreted by Parafollicular C-Cells in the **Thyroid Gland**. Calcitonin acts mainly on bone to Decrease plasma $[\text{Ca}^{++}]$

The actions and properties of each of these hormones will be discussed in detail.

Parathyroid Hormone



Parathyroid hormone (PTH) is secreted by the **Parathyroid Glands**, which adhere to or are imbedded within the posterior surface of the **Thyroid Glands**.

Humans typically have four parathyroid glands, which are flattened and elipsoid glands measuring about 6 mm in length and weighing less than 150 mg. They are well vascularized and derive their circulation from the inferior thyroid arteries. Parathyroid glands are comprised of two main cell types:

Chief Cells secrete **PTH** and are arranged in clusters typical of peptide hormone secreting cells. Chief cells are the predominant parathyroid cell type.

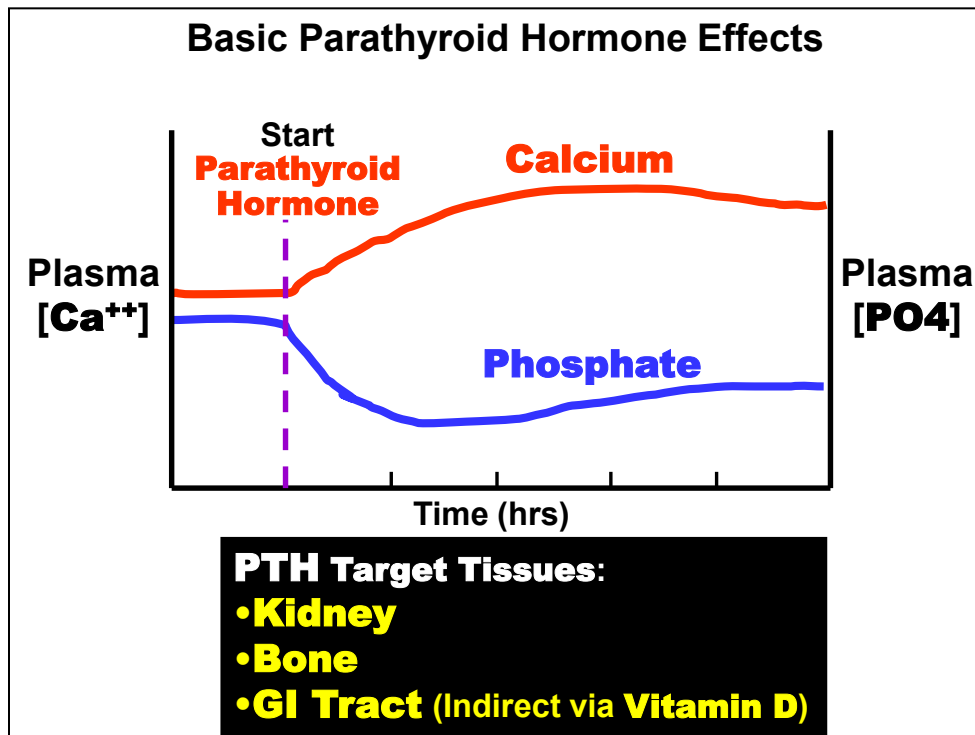
Oxyphil Cells have no known function, but may be a type of **Degenerating Cell**. They are larger than chief cells and appear singly or in small groups. Few oxyphil cells are observed prior to puberty but increase with age.

Parathyroid Hormone Properties

- **Classification:** Protein
- **Biosynthesis:** Parathyroid Chief Cells
- **Storage:** Vesicles
- **Secretion:** Exocytosis
- **Transport:** Dissolved
- **Metabolism:** Proteolysis (Half-Life: Minutes)
- **Mechanism:** Membrane Receptor: G-Protein
- **Actions:** Calcium & Phosphate Regulation
- **Regulation:** Negative Feedback: Plasma Ca^{++}
- **Disorders:** Excess: Hyperparathyroidism (e.g. Adenoma)
Deficiency: Hypoparathyroidism (e.g. Thyroidectomy)

Parathyroid Hormone endocrine properties

Highlights are summarized above.

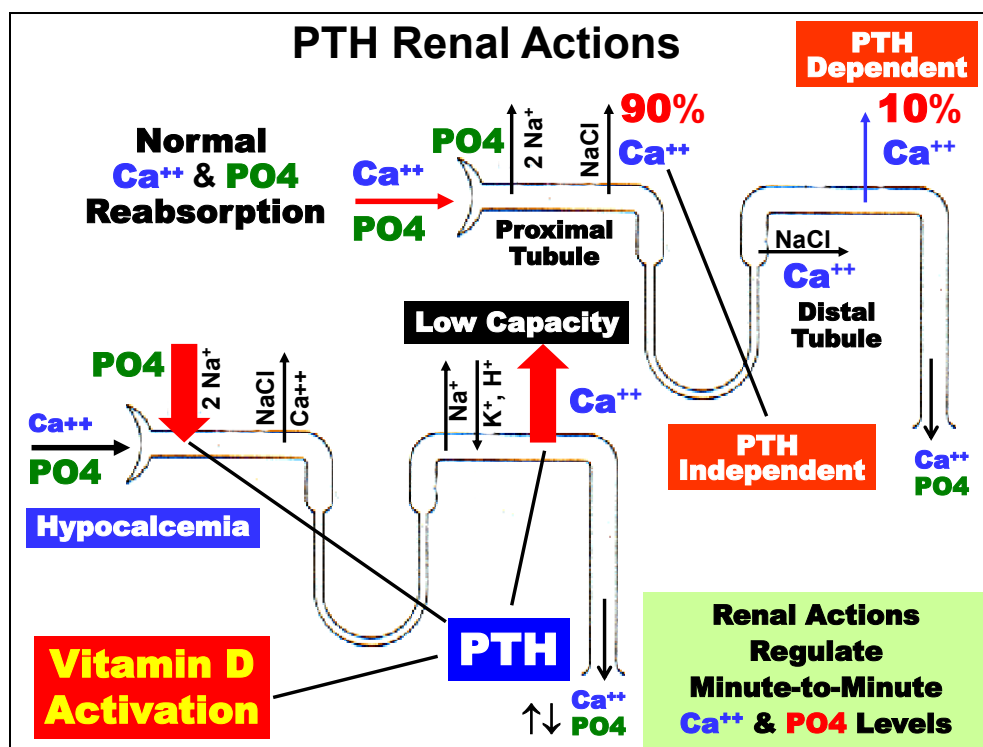


As discussed earlier, **PTH** is the principal regulator of **ECF $[Ca^{++}]$** . However, PTH actions also regulate of **ECF $[PO_3^-]$** even though its secretion is not directly regulated by phosphate.

(*graph*) As illustrated above, PTH actions have an **Inverse Effect** on plasma **$[Ca^{++}]$** (left y-axis) and **$[PO_3^-]$** (right y-axis) following the experimental infusion of PTH (dashed vertical line).

PTH exerts these effects through direct actions on the **Kidneys & Bone** as well as **indirect** actions on the **GI Tract** as a consequence of its regulation of **Vitamin D** activation (*discussed later*).

The next sections summarize the major physiological actions of PTH on the kidneys and in bone.



The kidneys respond **Rapidly** to changes in parathyroid hormone secretion and are primarily responsible for the **Minute-To-Minute Regulation** of ECF $[\text{Ca}^{++}]$. PTH has three distinct actions on the kidneys, which contribute to calcium homeostasis.

Calcium Reabsorption from Distal Nephron.

(top diagram) The illustration above represent a renal nephron. Note that $\approx 90\%$ of the total calcium reabsorbed by the kidneys occurs in the **Proximal Tubule**. This fraction of filtered Ca^{++} is **NOT** hormonally regulated and is, therefore, **PTH Independent**.

Only $\approx 10\%$ of filtered Ca^{++} reaches the Distal Tubule. The reabsorption of this fraction is regulated by PTH and is, therefore, **PTH Dependent**. Thus, a limitation of PTH stimulation of renal Ca^{++} reabsorption is that it has a **Low Capacity** that saturates when the filtered calcium load becomes too high. For example, only $\approx 10\%$ of the 10% of total filtered Ca^{++} that reaches the distal tubule can be reabsorbed even when PTH is maximally stimulated.

Phosphate Excretion from Proximal Tubule.

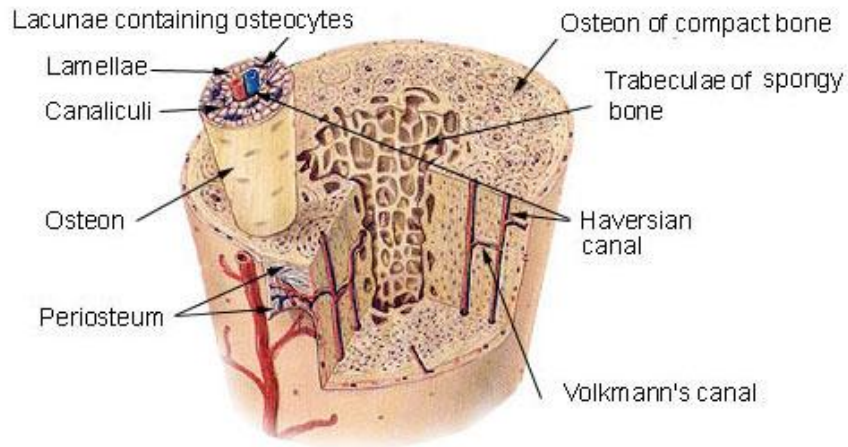
In contrast to Ca^{++} regulation, **PTH powerfully Inhibits tubular reabsorption of Phosphate (PO_4)** and thus **increases the urinary PO_4 Excretion**. This effect is exerted mainly at the **Proximal Tubule** where most PO_4 reabsorption occurs. PTH apparently decreases the abundance of **Na^+ - PO_4 Co-Transporters** along the proximal tubular cells by stimulating their translocation to intracellular stores.

Vitamin D Activation.

PTH stimulates the renal activation of Vitamin D, which is essential for the absorption of ingested Ca^{++} & PO_4 . This will be discussed in more detail later.

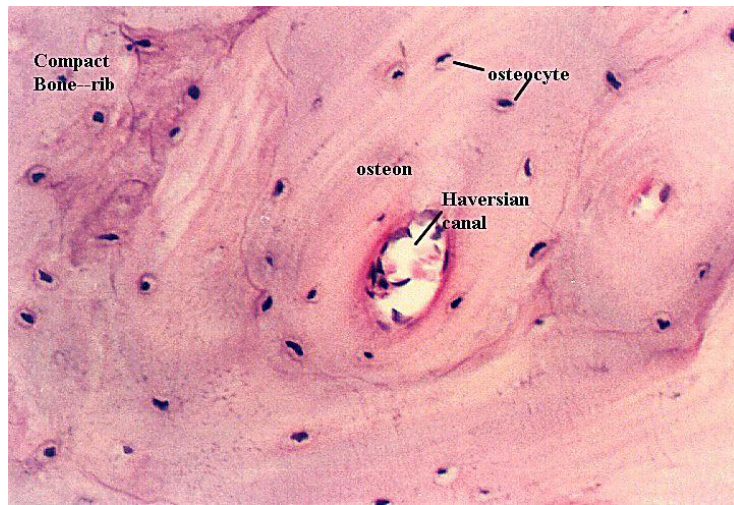
Bone-Basic Histology

Compact Bone & Spongy (Cancellous Bone)

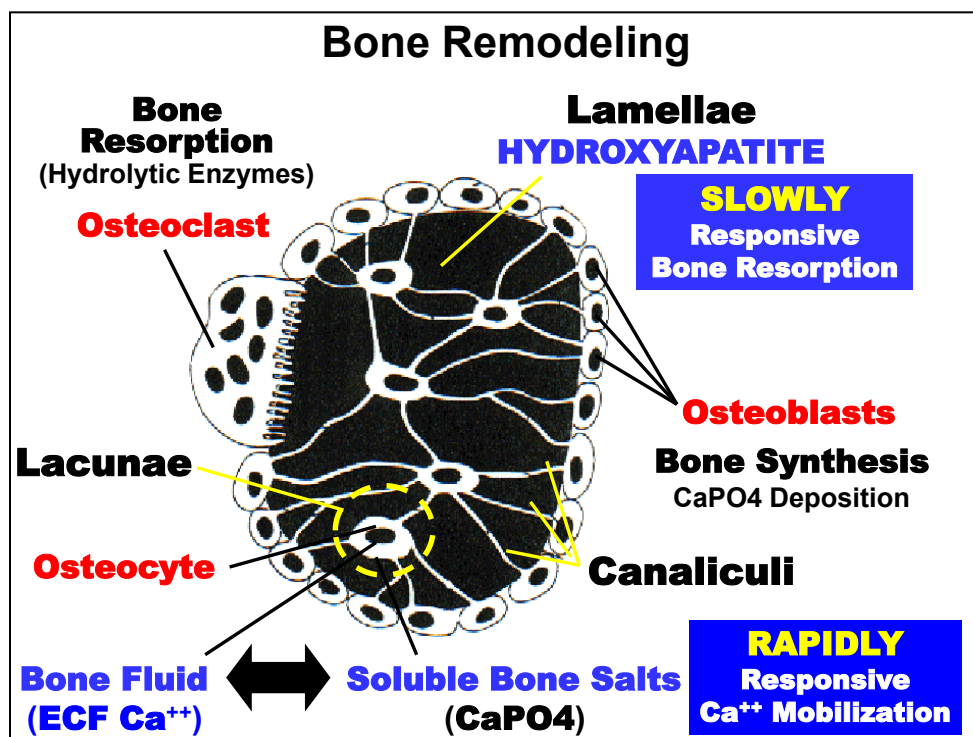


Review of bone histology, which is important for understanding the bone actions of PTH.

Compact Bone-Basic Histology



The **Osteon** is the functional unit of compact bone and the site of action of PTH, which regulates the functions of the various bone cells. These are summarized next.



In response to hypocalcemia, **Parathyroid Hormone stimulates Ca⁺⁺ mobilization from bone into ECF**. Ca⁺⁺ is exchanged between bone and ECF through two basic mechanisms.

Bone Fluid Exchange with Soluble Bone Salts.

Recall that **Lacunae** and **Canaliculi** (*bottom left*) are filled with **ECF (Bone Fluid Ca⁺⁺)** that is in equilibrium (*thick double arrow*) with **Soluble Bone-Salts** (CaPO₄) within the bone matrix (*black areas*). Bone Fluid is only separated from soluble Bone Salts by a membrane lining the lacunae.

Acute hypocalcemia ($\downarrow$ ECF [Ca⁺⁺]) causes the equilibrium to shift in the direction of Ca⁺⁺ mobilization via bone salt ionization. Because re-equilibration occurs quickly, calcium exchange between bone salts & bone fluid serves as a **Rapidly Responsive** mechanism that maintains ECF calcium homeostasis.

Bone Remodeling. **Hydroxyapatite** (*top center*) is continuously metabolized through a constant cycle of Degradation & Resynthesis termed **Bone Remodeling** that involves 3 bone cell types:

Osteoblasts (*middle right*) promote CaPO₄ deposition & mineralization into the hydroxyapatite and are primarily responsible for **Bone Synthesis**. As shown above, osteoblasts form a continuous sheet along the surface of newly forming bone within **Lamellae**.

Osteocytes (*bottom left*) are osteoblasts that become entrapped within **Lacunae** (*middle left*) when crystallized bone salts form around them. Long processes extend from osteocyte to osteocyte as well as with osteoblasts via **Canaliculi** (*bottom right*).

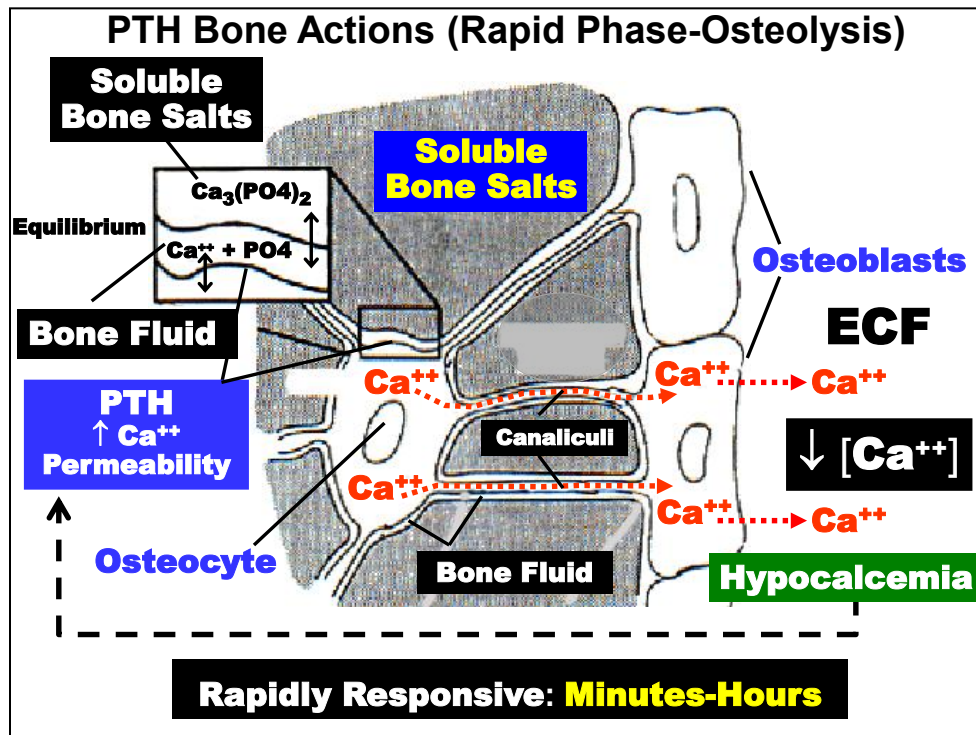
Osteoclasts (*top left*) secrete hydrolytic enzymes that degrade existing hydroxyapatite in a process termed **Bone Resorption**. Osteoclasts sweep over the surface of bone, dissolving protein matrix and crystals.

Normally, bone Resorption & Synthesis are balanced in healthy mature bone to maintain bone size.

However, under conditions such as **Chronic Hypocalcemia**, mobilization of soluble bone salts (*see above*) becomes limiting and Net Hydroxyapatite Resorption occurs in order to supply the needed Ca⁺⁺ to maintain normal ECF Ca⁺⁺ levels. Recall that bone integrity is always sacrificed to maintain the ECF [Ca⁺⁺].

Because of the extended time required to alter the remodeling equilibrium, this is a **Slowly Responding** mechanism that maintains ECF calcium homeostasis. The specific actions of PTH that result in these

mechanism is explained next.



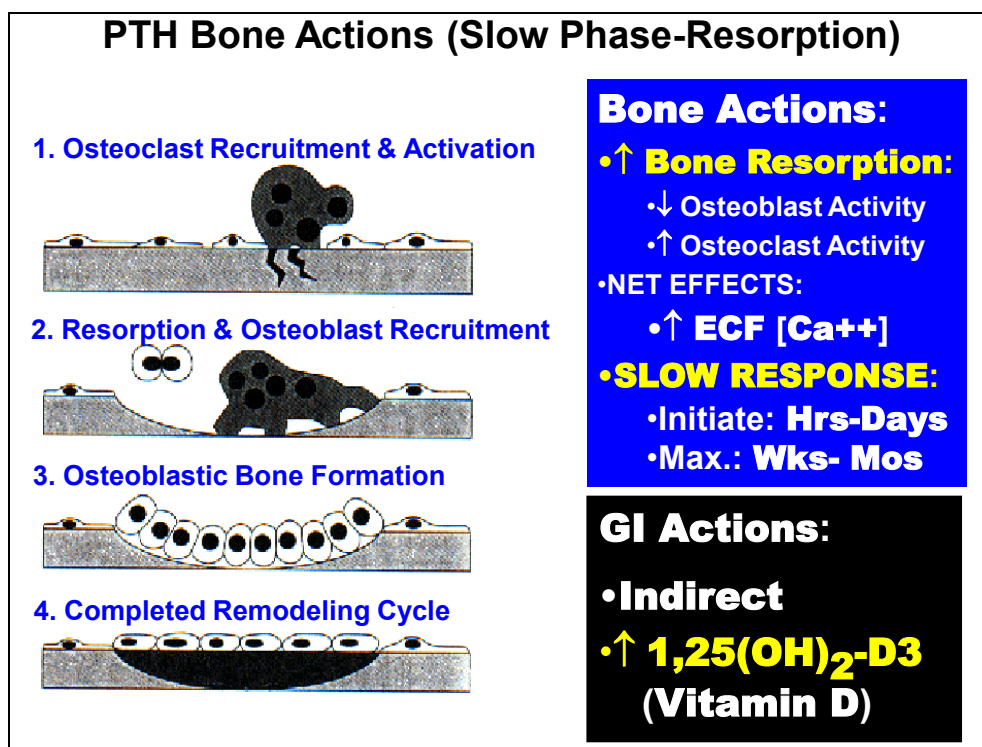
Parathyroid hormone actions are responsible for the **Rapid Phase** of Ca^{++} mobilization from bone, which begins within **Minutes** after the onset of hypocalcemia and can persist for **Hours**.

(left) PTH is thought to increase ($\uparrow$) **Ca^{++} Permeability** of the **Osteocyte** membrane that separates the **Bone Fluid (ECF)** from the **Soluble Bone Salt** compartment (*top left inset*).

This action increases Ca^{++} flux into osteocytes, driven by a steep concentration gradient from bone fluid into the cell. Increased Ca^{++} uptake by osteocytes lowers bone fluid $[\text{Ca}^{++}]$, which shifts the equilibrium toward **Soluble Bone Salt** ionization and Ca^{++} mobilization into **Bone Fluid** (*bottom center*).

This process is termed **OSTEOLYSIS**.

The Ca^{++} taken up by osteocytes is then transported via canaliculi leading to osteoblasts where it diffuses into the ECF (*dashed rightward arrows*) to normalize plasma $[\text{Ca}^{++}]$.

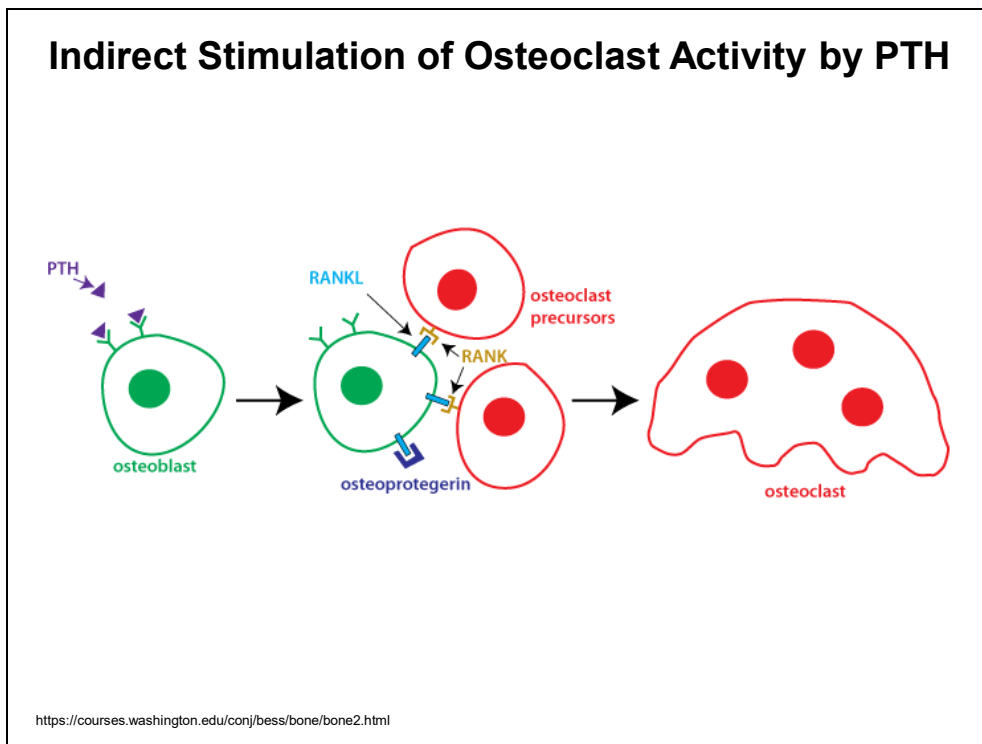


PTH also initiates the **Slow Phase of Ca⁺⁺ mobilization from bone** in response to hypocalcemia.

The steps in normal **Bone Remodeling** are summarized above (*left*). Recall the cooperative actions between Osteoclast (Resorption) & Osteoblast (Synthesis) on hydroxyapatite turnover. (*top right text box*) PTH increases Ca⁺⁺ mobilization from bone by causing net (↑) **Bone Resorption** through suppression of (↓) **Osteoblast Activity** & stimulation of (↑) **Osteoclast Activity**. Interestingly, osteoclasts do not possess PTH receptors and are induced indirectly via PTH stimulation of **RANKL** (receptor activator of nuclear factor-κB ligand) from osteoblasts, which basically allows formation of active osteoclasts. In any case, the net effect is **↑ ECF [Ca⁺⁺]**. Ca⁺⁺ enters the ECF via the same route described for the Rapid Phase.

Bone resorption is a potent mechanism for normalizing hypocalcemia. However, the response is relatively **SLOW** because of the delayed response of bone cells to PTH actions. Activation occurs only after many **Hours or Days** following the onset of hypocalcemia and **Weeks-Months** may be required for full develop.

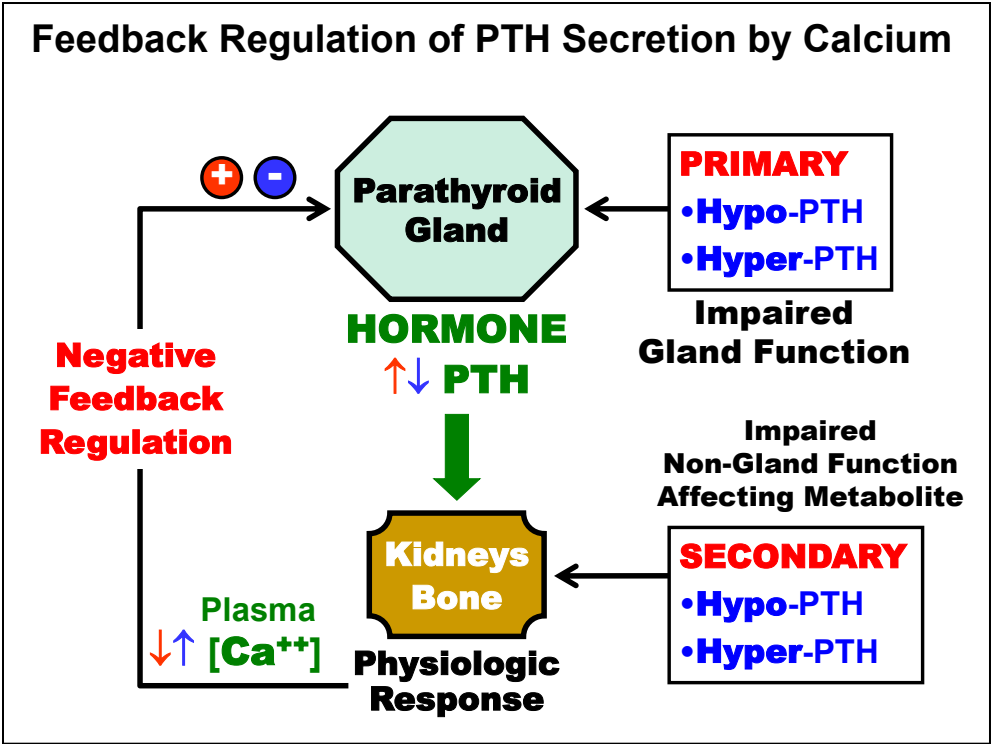
(*bottom right text box*) **PTH Actions on GI Tract**. Interestingly, PTH apparently has **NO** direct effects on either calcium or phosphate absorption from the GI tract. However, PTH acts to stimulate **Vitamin D** activation in the kidneys. Vitamin D actions and its relationship to PTH and calcium & phosphate homeostasis are discussed next.



PTH stimulates bone resorption by osteoclasts, but it does so indirectly. Receptors for PTH are located on osteoblasts, which then signal to bone marrow-derived osteoclast precursors to stimulate their fusion, differentiation and activation.

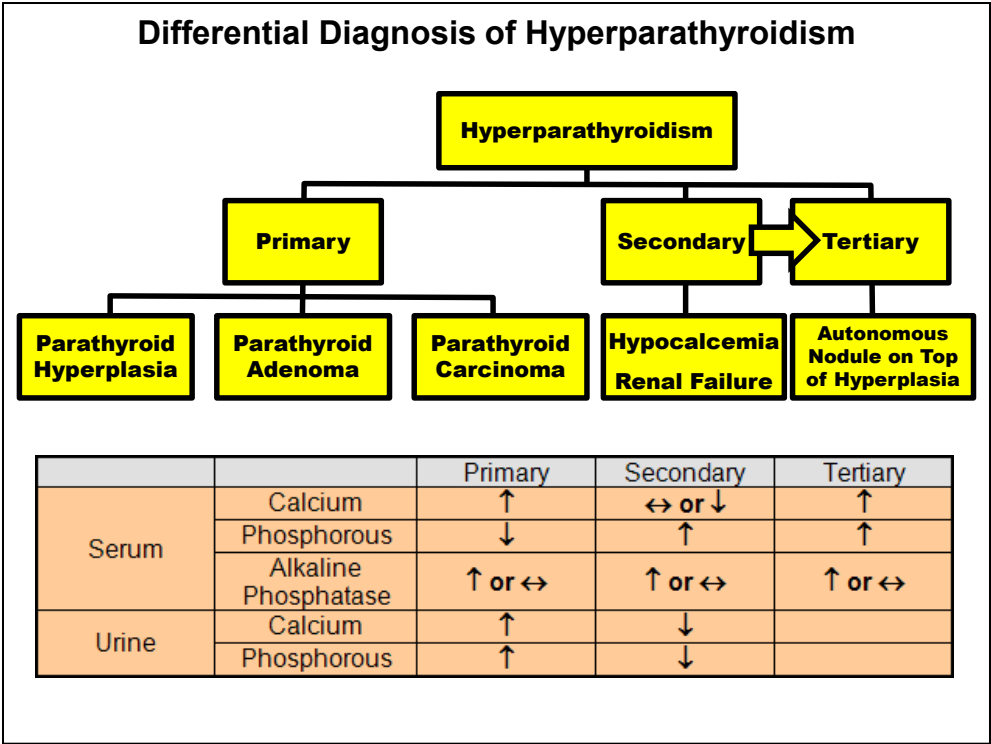
Osteoclast precursors express a cell-surface receptor known as **RANK** (Receptor Activator of Nuclear factor-Kappa B). Osteoblasts express **RANKL** (RANK Ligand) on the extracellular surface of their plasma membrane.

When they are stimulated by PTH, osteoblasts up-regulate expression of RANKL, which binds to RANK, activating signaling pathways that promote osteoclast differentiation and survival. Osteoblasts also express a secreted factor called **Osteoprotegerin**. As its name implies, osteoprotegerin "protects bone" by preventing bone resorption. Osteoprotegerin works as a decoy receptor for RANKL: it binds RANKL and therefore prevents binding to RANK and stimulation of osteoclastogenesis. The ratio of osteoprotegerin:RANKL produced by osteoblasts will determine the extent of bone resorption.



Feedback regulation of PTH secretion by plasma calcium and disorder classification.

Parathyroid Hormone Disorders



Summary of differential laboratory diagnosis of **Hyperparathyroidism**.

Primary Hyperparathyroidism

1° Hyperparathyroidism:

•e.g. **Parathyroid Adenoma**

•Hypercalcemia:

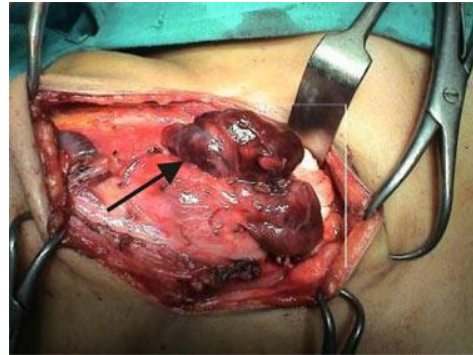
- Muscle Weakness, Fatigue
- Bradycardia
- Constipation, Nausea
- CNS Depression

•Bone Effects:

- Net Bone Resorption
- Weakened Bone
- Bone Pain

•Renal Effects:

- Kidney Stones



Laboratory Diagnosis:

- ↑ **PTH**
- ↑ Serum **Ca⁺⁺**
- ↓ Serum **PO₄**
- ↑ **Alkaline Phosphatase**

Primary Hyperparathyroidism: Basic disorder is **Excess PTH** secretion due to parathyroid gland dysfunction. **Parathyroid Adenoma** (*top right*) accounts for **80-85%** of cases. Most clinical features related to primary development of **Hypercalcemia**.

Hypercalcemia Effects. Hypercalcemia causes Skeletal Muscle Weakness & Fatigue as well as abnormal electrical activity of the heart (e.g., Bradycardia). Contractile impairment suppresses GI motility, resulting in Nausea, Constipation & Lack of Appetite. Impaired muscular contraction is caused by **decreased membrane Na⁺ permeability** that decreases depolarization sensitivity (i.e., hyperpolarization due to increased cation charge. Decreased **Na⁺** permeability, is attributed to **increased Ca⁺⁺** binding to Na⁺ channels, which increases positive charge on the channel and decreases its sensitivity to voltage changes and results in hyperpolarization. As discussed later, hypocalcemia causes the opposite effects on muscle contractile state (i.e., **Tetany**). Sluggish neural activity can cause development of **CNS Depression**.

Bone Effects. Enhanced Bone Remodeling, but Net Bone Resorption due to enhanced Osteoclast activity. Results in greatly Weakened Bones with severe Bone Pain (*osteitis fibrosis cystica*) that greatly increases the potential for bone fracture.

Renal Effects. Mild hyperparathyroid patients commonly develop **Kidney Stones**, which are calcium phosphate crystals that precipitate in the kidneys as a result of the proportionate increase in renal calcium and phosphate filtration.

Laboratory Diagnosis. Typical laboratory findings diagnostic of primary hyperparathyroidism are summarized above (*bottom right text box*). Rationalize the findings based on the known physiological actions of PTH. What is the role of alkaline phosphatase and why would it be elevated?

Secondary Hyperparathyroidism

	PTH	Ca ²⁺	PO ₄ ³⁻	1,25-hydroxyvitamin D
Renal Failure	↑	↓	↑	↓
Hypovitaminosis D	↑	↓	↓	↓
Hypocalcemia	↑	↓	↓	↑
Hyperphosphatemia	↑	↑	↑	↓

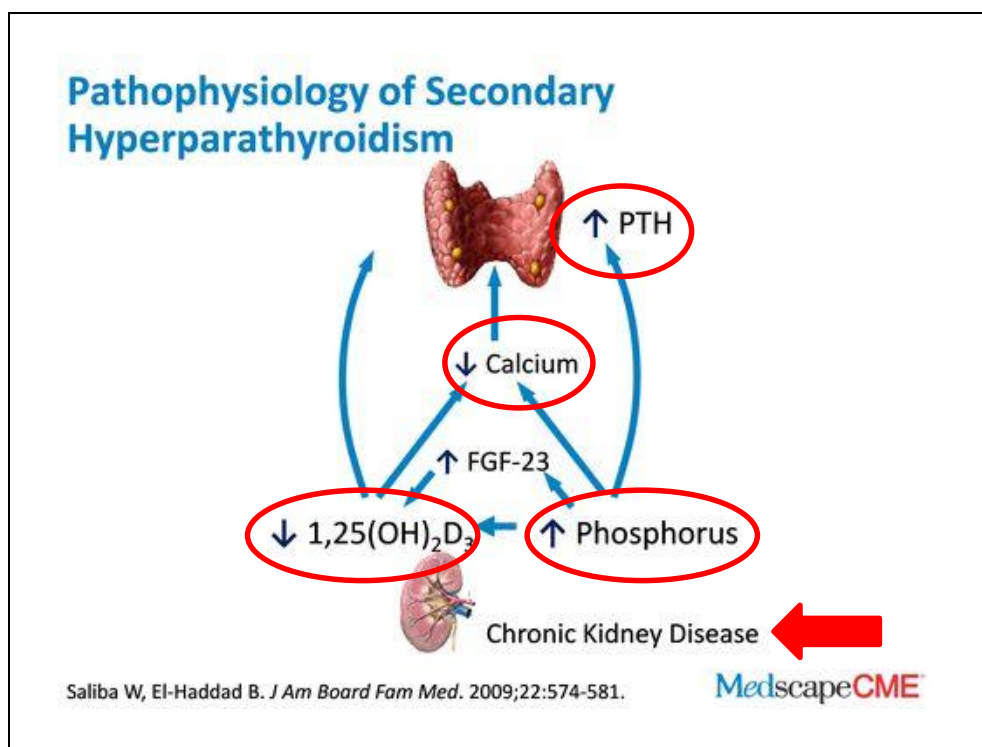
<https://www.youtube.com/watch?v=yYWgdCga5wk>

Common Etiologies of Secondary Hyperparathyroidism.

Try to rationalize pathophysiological outcomes summarized in Table and compare and contrast with typical lab values associated with **Primary Hyperparathyroidism**.

Review the Youtube video to evaluate your rationale:

<https://www.youtube.com/watch?v=yYWgdCga5wk>



Example of rationalizing the pathophysiology of **Secondary Hyperparathyroidism...Chronic Kidney Failure**

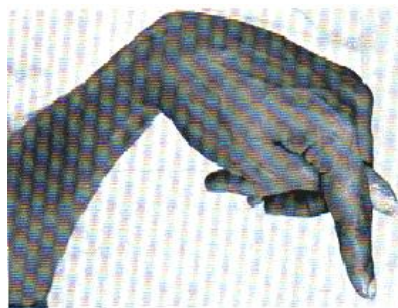
1. Decreased renal calcium reabsorption leads to low serum Ca⁺⁺
2. Decreased renal phosphate excretion leads to high serum phosphate, which worsens hypocalcemia through increased CaPO₄ formation and suppresses vitamin D activation.
3. Decreased renal vitamin D activation leads to deficiency that exacerbates hypocalcemia
4. All outcomes lead to compensatory hyperPTH secretion.

FGF-23 effects not discussed.

Primary Hypoparathyroidism

1° Hypoparathyroidism:

- e.g. **Thyroidectomy**
- **Hypocalcemia**
- Mostly Neuromuscular Symptoms:
- **TETANY**
 - ↓ **Ca⁺⁺ Binding to Na⁺ Channels**
 - ↑ **Na⁺ Membrane Permeability**
 - ↑ **Membrane Depolarization**



Trousseau's Sign

Laboratory Diagnosis:

- ↓ **PTH**
- ↓ **Serum Ca⁺⁺**
- ↑ **Serum PO₄**

Primary Hypoparathyroidism Basic disorder is **Deficient PTH** secretion due to parathyroid gland dysfunction. In contrast to hyperparathyroidism, hypoparathyroidism is rare (incidence?). Most common causes is surgical removal (e.g. **Thyroidectomy**). Most clinical features related to primary development of **Hypocalcemia**, which manifests as **Neuromuscular Impairments**.

Clinically, the hallmark of severe hypoparathyroidism (hypocalcemia) is **TETANY**. This state of spontaneous tonic muscular contraction is caused by **increased membrane Na⁺ permeability** that increases depolarization sensitivity. Increased **Na⁺ permeability**, is thought to be the result of **decreased Ca⁺⁺ binding to Na⁺ channels**, which apparently decreases the positive charge on the channel and increases its sensitivity to voltage changes.

Involuntary tetanic muscle contractions are painful. The hands are typically involved but other muscle groups are affected. The principal manifestations in the hands usually begins with adduction of the thumb, followed by flexion of the wrist, forming the characteristic "**TROUSSEAU'S SIGN**" (*bottom left*), which can be elicited clinically in patients as diagnostic of the condition.

Laboratory Diagnosis. Typical laboratory findings diagnostic of primary hypoparathyroidism are summarized above (*bottom right text box*). Rationalize the findings based on the known physiological actions of PTH.

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Vitamin D

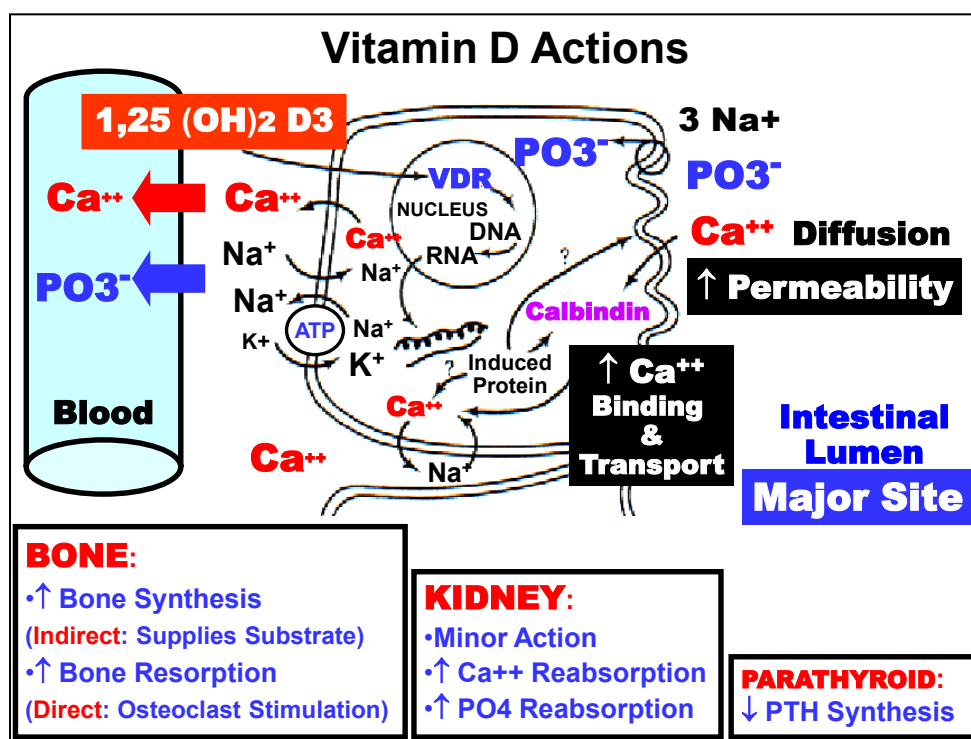
**1,25 (OH)₂ Cholecalciferol
Calcitriol**

Vitamin D Hormone Properties

- **Classification:** Steroid (Cholesterol)
- **Biosynthesis:** Skin-Liver-Kidney
- **Storage:** No
- **Secretion:** Passive Diffusion
- **Transport:** Bound
- **Metabolism:** Modification (Long Half-Life: Hours-Days)
- **Mechanism:** Intracellular Membrane Receptor
- **Actions:** GI Calcium & Phosphate Absorption
- **Regulation:** Activation (Plasma Ca^{++})
- **Disorders:** Excess: Rare
Deficiency: Rickets/Osteomalacia

Vitamin D endocrine properties

Highlights are summarized above.



The net effects of 1,25(OH)₂D₃ actions are to **increase [Ca⁺⁺] & [PO₄]** in ECF. The major site of action is the **Small Intestine**. However, important actions are exerted in **Bone & Kidney** and 1,25(OH)₂D₃ also plays a regulatory role in PTH secretion.

Intestinal Actions. As illustrated above, 1,25(OH)₂D₃ stimulates absorption of dietary calcium (Ca⁺⁺) & phosphate (PO₃⁻) from the small intestine.

Calcium Absorption. Ca⁺⁺ entry into enterocytes (*top right*) occurs by diffusion down its own concentration gradient. Ca⁺⁺ uptake into blood (*left*) depends on a basolateral 2° Active Transport mechanism involving a Na⁺-Ca⁺⁺ Exchanger that is dependent upon energy (ATP) supplied by membrane Na⁺-K⁺ ATPase activity.

Although details remain unclear, 1,25(OH)₂D₃ may increase Ca⁺⁺ entry from the lumen by **↑ Ca⁺⁺ Permeability** of the apical brush border membrane. In addition, 1,25(OH)₂D₃ also stimulates the synthesis of several Vitamin D Induced Proteins including, **CALBINDIN**, which may function to **Bind Ca⁺⁺** after diffusing into enterocytes from the lumen. This keeps cytosolic [Ca⁺⁺] LOW, maintaining a steep gradient for Ca⁺⁺ influx. Calbindin may also act as a shuttle to Transport Ca⁺⁺ to basolateral Na⁺-Ca⁺⁺ exchangers.

Phosphate Absorption Although Ca⁺⁺ uptake is generally accompanied by phosphate (PO₃⁻) uptake, PO₃⁻ is absorbed by independent mechanisms. PO₃⁻ entry from the lumen occurs through a Na⁺-dependent apical transporter. Basolateral uptake of PO₃⁻ into blood is unclear. Moreover, the mechanisms by which 1,25(OH)₂D₃ increases PO₃⁻ transport is not well understood.

Bone Actions. 1,25(OH)₂D₃ promotes **Bone Resorption**.

Bone Synthesis. 1,25(OH)₂D₃ apparently does **NOT** directly stimulate bone formation, but only acts indirectly through increased Ca⁺⁺ & PO₃⁻ absorption (Supplies Substrate).

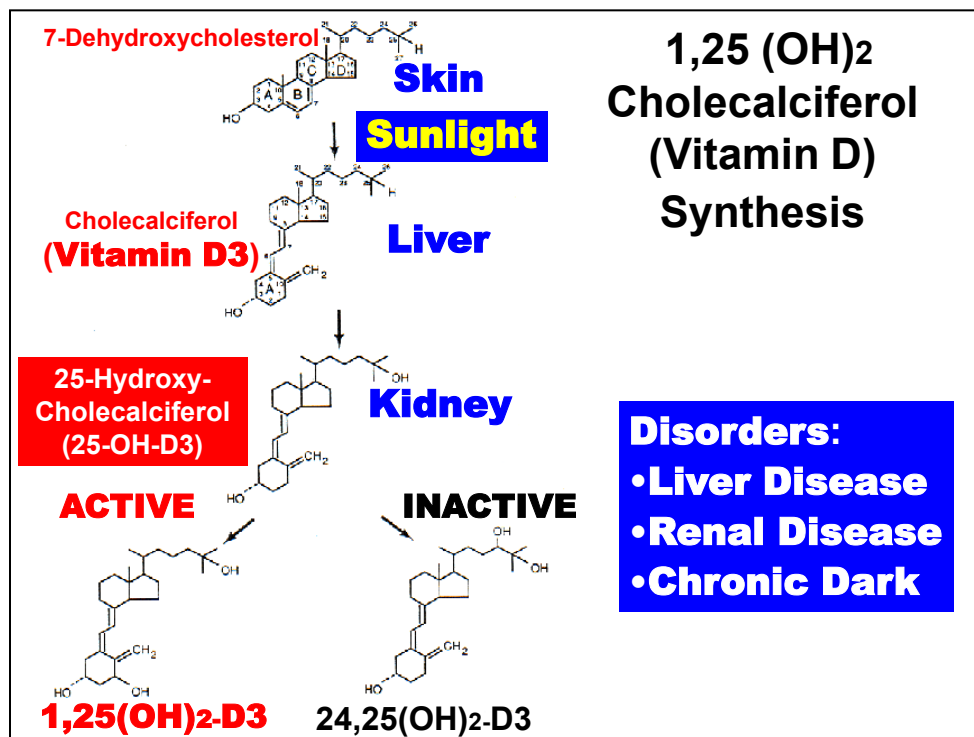
Bone Resorption. 1,25(OH)₂D₃ directly **↑ Bone Resorption** through Osteoclast Stimulation.

Kidney Actions. 1,25(OH)₂D₃ increases renal reabsorption of **BOTH Ca⁺⁺ & PO₄**.

Phosphate Reabsorption. Effect of 1,25(OH)₂D₃ on PO₃⁻ reabsorption is probably indirect, occurring as a consequence of its inhibition of PTH secretion (*below*).

Calcium Reabsorption. Effect of 1,25(OH)₂D₃ on renal Ca⁺⁺ reabsorption is probably direct 1,25(OH)₂D₃ receptors are found in the distal tubule where selective reabsorption of Ca⁺⁺ occurs. However, regulation is weak relative to PTH.

Parathyroid Gland Actions. 1,25(OH)₂D₃ **suppresses PTH**.



The active form of **Vitamin D**, **1,25 Dihydroxycholecalciferol (1,25(OH)₂D₃)** is essential for acquiring adequate amounts of calcium & phosphate from the diet.

1,25(OH)₂D₃ is classified as a **Steroid (Sterol) Hormone**. However, in contrast to classical cholesterol derived steroid hormones like glucocorticoids, the immediate precursor is the molecule **7-Dehydroxycholesterol**, which is actually synthesized in the **SKIN** from Acetyl CoA and is considered the **STORAGE FORM** of active Vitamin D.

(left) Irradiation of the skin (e.g. **Sunlight**) results in the opening of the B ring of the steroid nucleus and the formation of **Vitamin D₃ (Cholecalciferol)**. Conversion is important because Cholecalciferol has a high affinity for a vitamin D-binding globulin that allows solubilization and transport in blood. 7-Dehydroxycholesterol is unable to bind.

Cholecalciferol is transported in the blood to the **LIVER** where it is oxidized to form **25-Hydroxycholecalciferol (25-OH-D₃)**, which is the major circulating form of vitamin D but possesses only weak biological activity.

25-OH-D₃ is further oxidized after transport to the **KIDNEY** to yield the **ACTIVE** Vitamin D product: **1,25-Dihydroxycholecalciferol (1,25(OH)₂D₃)**, which accounts for almost all of the biological activity of Vitamin D.

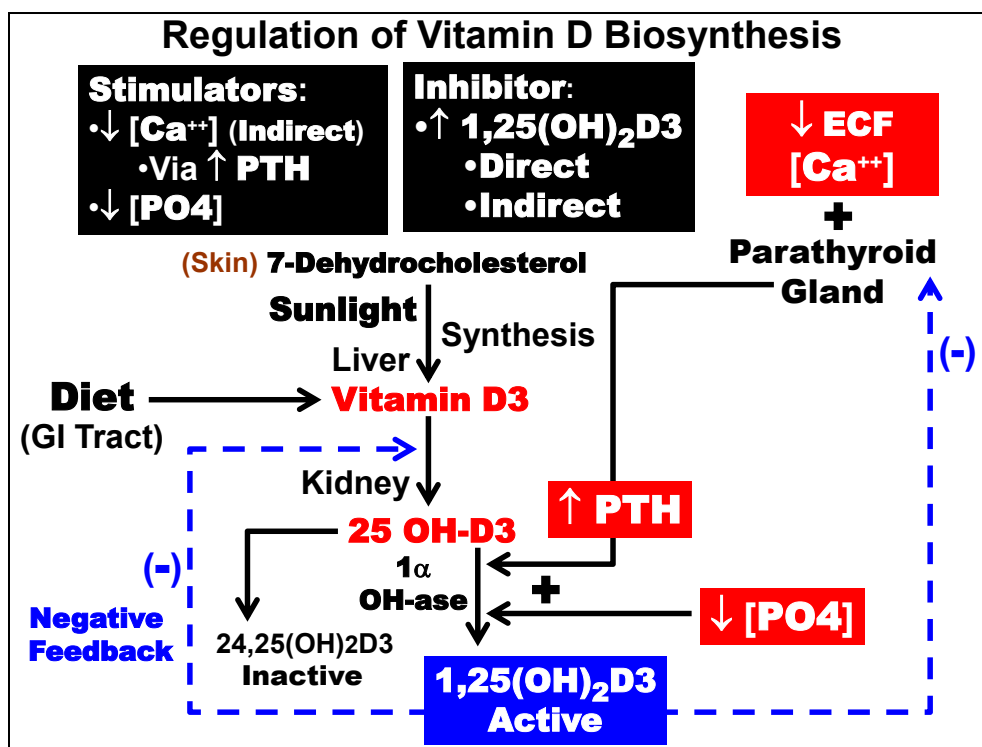
Thus, an important concept is that the **Biologically Active Form of Vitamin D is Secreted from the Kidneys**.

CLINICAL NOTE: (middle right text box) Unlike most hormones, Vitamin D synthesis and activation require the coordination of enzymatic actions performed by several different organs (skin, liver, kidney). Thus, **Liver & Kidney Disease** and/or prolonged absence of sunlight (**Chronic Dark**) commonly cause impaired Vitamin D synthesis and/or metabolism, with consequential impairment of its actions.

Note that 25-hydroxycholecalciferol can also be alternatively converted to the compound **24,25-dihydroxycholecalciferol (24,25(OH)₂D₃)** in the kidneys. Although about 100x more abundant than 1,25(OH)₂D₃, 24,25(OH)₂D₃ has little biological activity due to its **low affinity** for Vitamin D receptors in intestinal mucosal cells. It appears that 24,25(OH)₂D₃ is likely the initial metabolite in the degradative or inactivation pathway of Vitamin D. Subsequent reactions include oxidation of the side chain and biliary excretion.

Other than the distinctions cited above, 1,25(OH)₂D₃ possesses the Basic Properties Typical of Steroid

Hormones.



The major step in vitamin D production is the hydroxylation of **25(OH)-D₃** by the enzyme **1α-Hydroxylase (1α OH-ase)** found in **kidney proximal tubule** cells, which forms **Active 1,25(OH)₂D₃**. As summarized above, this process is subject to + & - regulation by a number of factors.

Stimulatory Factors (top left text box)

Low ECF [Ca⁺⁺] (Hypocalcemia) stimulates active vitamin D biosynthesis. However, 1,25(OH)₂D₃ activation is NOT directly stimulated by hypocalcemia.

This effect occurs indirectly through **Parathyroid Hormone** induction. Thus, PTH is the major direct stimulator of vitamin D activation. PTH primarily acts by increasing **1α-Hydroxylase** synthesis. **Low ECF [PO₄]** stimulation of vitamin D production also occurs directly. However, physiological stimulation probably occurs mainly as an indirect consequence of hypophosphatemia resulting from PTH stimulated renal PO₃- secretion.

Inhibitory Factors (top right text box)

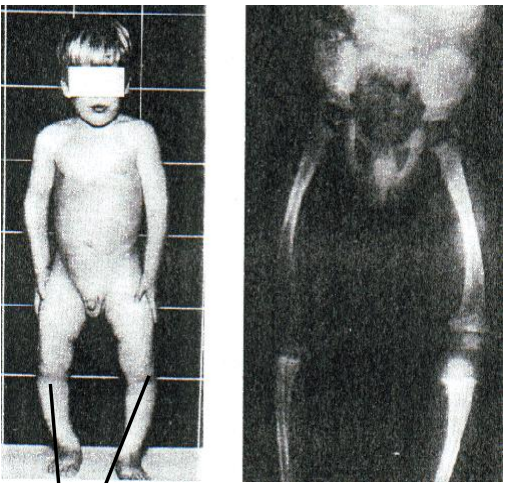
1,25(OH)₂D₃ acts as a negative feedback inhibitor of its own production through actions at several steps in the biochemical process (*left dashed arrow*). These actions both decrease production of 1,25(OH)₂D₃ and divert precursor compounds toward the inactivation or degradative pathways e.g. **24,25(OH)₂D₃** production.

(*right dashed arrow*) As indicated, **1,25(OH)₂D₃** directly inhibits PTH. In addition, the net effects of 1,25(OH)₂D₃ action (↑ ECF Ca⁺⁺ & PO₄) both act to suppress PTH & [PO₄] in a classical negative feedback manner.

Rickets - Osteomalacia

Vitamin D Deficiency

- **RICKETS:** Children
- **OSTEOMALACIA:** Adults
- **Clinical Features:**
 - Bone Loss/Weakness
 - Rarely Tetany
- **Children**
 - Short Stature
 - Bowed Legs
- **Laboratory Diagnosis:**
 - ↓ **1, 25 (OH)₂-D₃**
 - ↑ **PTH (2° Hyper-PTH)**
 - ↓ or ↔ Serum [Ca⁺⁺]
 - ↓ ↓ Serum [PO₄]
 - ↑ Alkaline Phosphatase



RICKETS

Insufficient Calcification

Rickets is a condition mainly observed in **children** and is most commonly caused by a **deficiency in Vitamin D production & activation**, availability in the diet, or action at target cells. **Osteomalacia** is the development of rickets in adults and is frequently called "Adult Rickets". Lack of sun exposure and dietary deficiencies in adults occur, but are rare. However, serious deficiencies of vitamin D and calcium occasionally occur as a result of chronic steatorrhea (fat malabsorption), which causes fat soluble vitamin D and calcium to pass unabsorbed in the feces.

Clinical Features

Bone Effects. In adults & children, **Bone Loss** (Osteoporosis) and **Bone Weakness** are the most common manifestations. In children, Rickets produces **Short Stature** because of lack of calcium & phosphate substrate for growing long bones. The situation is aggravated by the bone resorption actions of PTH in preventing hypocalcemia during chronic vitamin D deficiencies. Recall that PTH vigorously protects against hypocalcemia by increasing Osteoclast activity. Although Osteoblast activity also increases to prevent net bone loss, the osteoid that is produced is inadequately calcified because of insufficient calcium & phosphate availability. Consequently, the newly formed, uncalcified and weak osteoid gradually takes the place of the resorbed calcified bone. This generally causes bone pain and tenderness and the development of the "**Bowed Legs**" characteristic of children with Rickets (*see photos above*).

Tetany is not usually observed in the early stages of rickets due to increased osteoclastic activity on bone, which maintains ECF [Ca⁺⁺]. Adults who develop osteomalacia almost never develop tetany. However, after exhaustion of bone reserves, [Ca⁺⁺] may fall quickly. As blood calcium levels approach ≈7 mg/dL, the usual signs of tetany develop and the child may succumb to tetanic respiratory spasm unless intravenously treated with exogenous calcium.

Treatment of Rickets. Treatment is dependent on supplying both adequate amounts of calcium & phosphate in the diet AND administering adequate amounts of vitamin D. If vitamin D is not administered along with the calcium, little calcium and phosphate will be absorbed from the

intestine.

Rickets (con't)

Laboratory Diagnosis.

In addition to **decreased 1, 25 (OH)²-D₃** levels, **PTH** is **elevated** as a compensatory response to hypocalcemia (**2° Hyperparathyroidism**).

Hypocalcemia or Eucalcemia. Plasma [**Ca⁺⁺**] is usually only slightly depressed (↓) or even normal (↔) because **PTH** is capable of maintaining calcium homeostasis through increased bone resorption and renal **Ca⁺⁺** reabsorption.

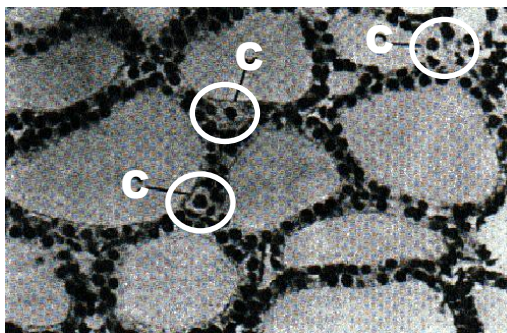
Severe Hypophosphatemia. By contrast, plasma [**PO₄**] is greatly depressed (↓↓) primarily because compensatory increases in **PTH** activity in response to hypocalcemia causes increased phosphate excretion from the proximal tubules. This underscores the relatively weak regulatory system that maintains phosphate homeostasis.

The latter two manifestations assist in distinguishing this condition from Primary Hyperparathyroidism. Also see laboratory differential diagnosis section of Hypothalamus-Pituitary section.

Alkaline Phosphatase may also become **elevated** as a consequence of increased bone resorption and turnover.

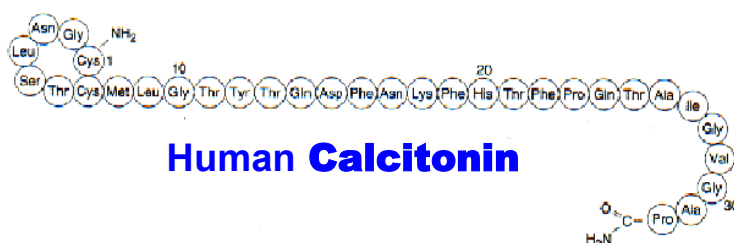
Calcitonin

Calcitonin: Basic Properties



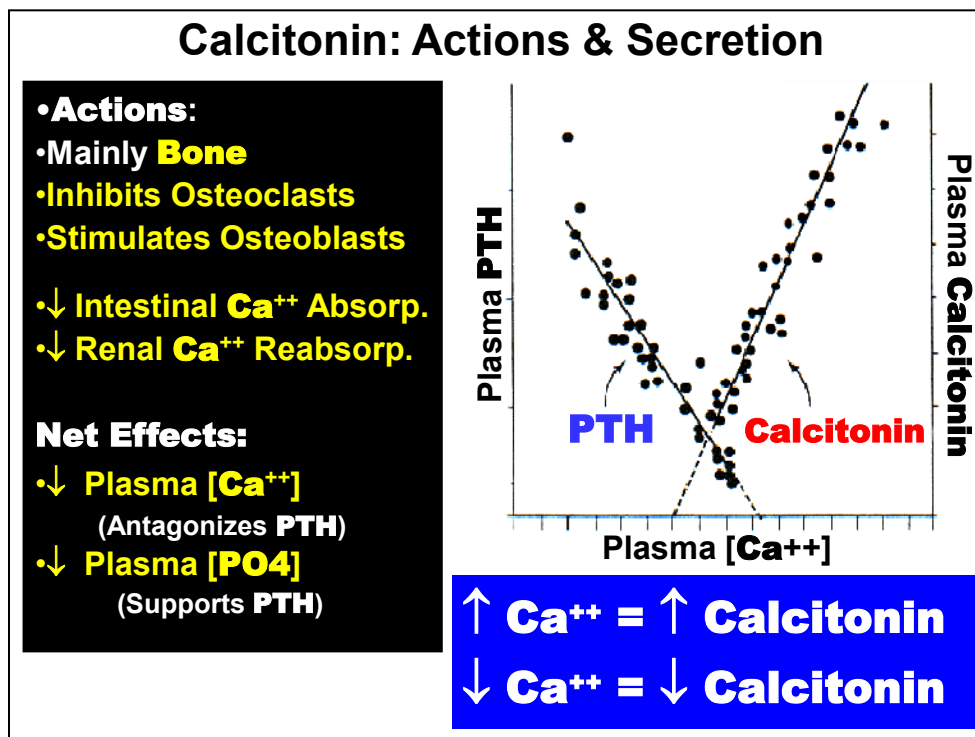
**Basic
Properties
Typical of
Protein Hormone**

Thyroid Gland: Parafollicular (C) Cells



Calcitonin is a **Peptide Hormone** (32 amino acids; MW 3400) secreted from **Thyroid Gland Parafollicular (C) Cells**. These cells, also called simply **C cells**, occur singly or in clusters in or between thyroid follicles. The primary structure is shown above.

The Basic Properties of Calcitonin are Typical of Protein Hormones.

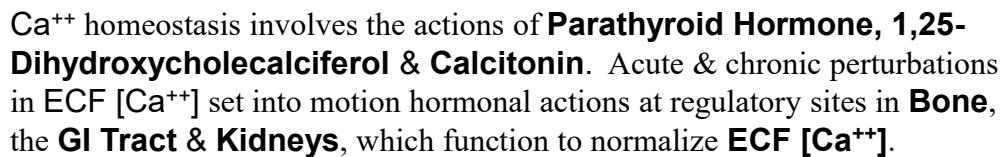


Calcitonin has been shown experimentally to cause a rapid and dramatic decrease in plasma $[\text{Ca}^{++}]$ and $[\text{PO}_4]$. However, calcitonin apparently is **NOT** a major physiological regulator of calcium or phosphate homeostasis in humans. The importance of calcitonin in humans may be limited to protection against excessive bone resorption (*see below*). In this regard, calcitonin is sometimes used pharmacologically to treat calcium related disorders such as **Osteoporosis**.

Calcitonin exerts its major actions on **Bone**. Calcitonin lowers blood calcium and phosphate primarily by **Inhibiting Osteoclast** activity, but also **Stimulates Osteoblasts**. Secretion is highest when osteoclastic bone resorption is high and lowest when the resorptive rate is low.

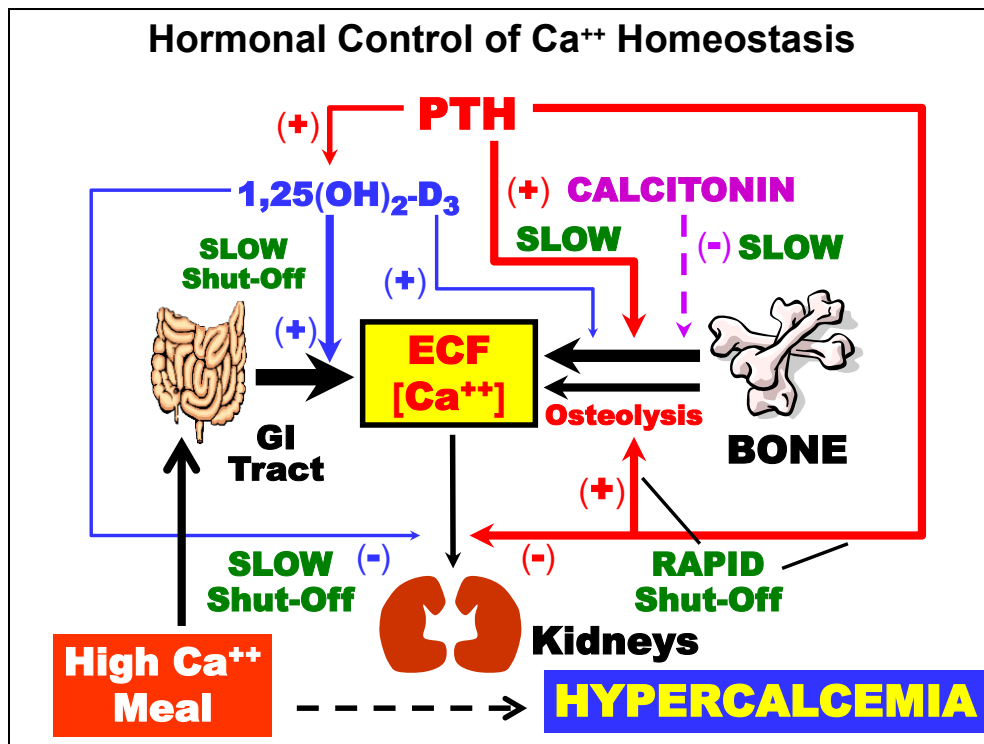
In addition, calcitonin also **decreases intestinal Ca^{++} absorption and renal tubule Ca^{++} reabsorption**.

As indicated above (*right graph*), **thyroid parafollicular C cells** secrete **Calcitonin** (*right y-axis*) in direct response to an increase in plasma $[\text{Ca}^{++}]$ (*x-axis*). Interestingly, parafollicular cells possess the same calcium sensing receptors as described earlier for PTH chief cells. Thus, PTH (*left y-axis*) and calcitonin both respond over the same Plasma $[\text{Ca}^{++}]$ range, but their secretory responses are opposite, presumably because of differences in the secretory apparatus response.



Rapid Responses (*outer right arrows*). **PTH** secretion (*top*) occurs in response to even very small decreases in $[Ca^{++}]$. Rapid **PTH** actions include (1) Inhibition (-) of **Renal Ca^{++} Excretion** from the distal tubules & (2) Stimulation (+) of **Bone Osteolysis** (soluble bone salt). Some renal augmentation by **1,25(OH)2D3**. These actions are initiated within **Minutes-Hours** after onset of hypocalcemia, but are usually only able to compensate for mild & brief hypocalcemia episodes.

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HYPERCALCEMIA is rarely observed pathophysiologically. However, one possible scenario involves an individual who has subsisted on a prolonged **Low Calcium Diet** and then suddenly ingests a **High Ca^{++} Meal** as depicted above.

Prior to this meal, **Rapid & Slow Responses to Chronic Hypocalcemia** have resulted in HIGH **PTH** & **1,25(OH) $_2$ D $_3$** levels, causing elevated **Bone Osteolysis & Resorption** and maximal **Intestinal Ca^{++} Absorption & Renal Ca^{++} Reabsorption**. When the high calcium meal is consumed under these circumstances, **ECF $[\text{Ca}^{++}]$** rapidly increases (**Hypercalcemia**), initiating the following compensatory regulatory processes:

Rapid Suppression. **PTH** secretion decreases immediately, increasing **Renal Ca^{++} Excretion** & Decreasing **Bone Osteolysis**.

Slow Suppression. In contrast to rapid **PTH** actions, Several **Hours** are required for **PTH** stimulated renal **1,25(OH) $_2$ D $_3$** production and osteoclast activity to diminish. Many more hours (**10-20 hours**) are required to suppress **1,25(OH) $_2$ D $_3$** actions on all tissues due to its relatively long half-life.

Much later, **Calcitonin** secretion (*top right*) may be necessary to directly **Inhibit Osteoclast Activity**, which more rapidly slows **Ca^{++} influx** into the **ECF**.

Narrated PowerPoint Presentations

- 1. Overview & Principles of Endocrine System**
- 2. Hypothalamic Neuroendocrine Hormones**
- 3. Anterior Pituitary Hormones**
- 4. Posterior Pituitary Hormones**
- 5. Thyroid Hormones Basic Properties**
- 6. Thyroid Hormone Disorders**
- 7. Parathyroid Hormone**
- 8. Vitamin D and Calcitonin**
- 9. Adrenal Cortex Hormones**
- 10. Steroid Hormone Biosynthesis & CAH Disorders**
- 11. Adrenal Medulla Hormones**